

Practical 9: Forecast of commercial loans

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1. Question

Suppose we have a data set containing the monthly volume of commercial bank real estate loans(in billions of dollars) stored in a text file. Write the R code to forecast the volume of commercial loans for next 20 months.

2. Objective

The primary goal of this analysis is to quantify and project the future trajectory of commercial bank real estate loans based on historical monthly volume data.

3. Code

```
# Set working directory (edit path as needed)
setwd("C:\\Users\\Sandeep Balaji\\Downloads\\TSA-LAB-main\\TSA-LAB-main")

# Import data from a text file
raw_data <- read.table("bank_case.txt", header = FALSE)

# 3. Convert to a Time Series (ts) object
# Change 'start' and 'frequency' based on your data's actual start date
# frequency = 12 indicates monthly data
loan_ts <- ts(raw_data$V1, start = c(2020, 1), frequency = 12)

# 4. Fit an automated forecasting model (Auto-ARIMA)
# auto.arima automatically selects the best parameters for trend and seasonality
fit <- auto.arima(loan_ts)

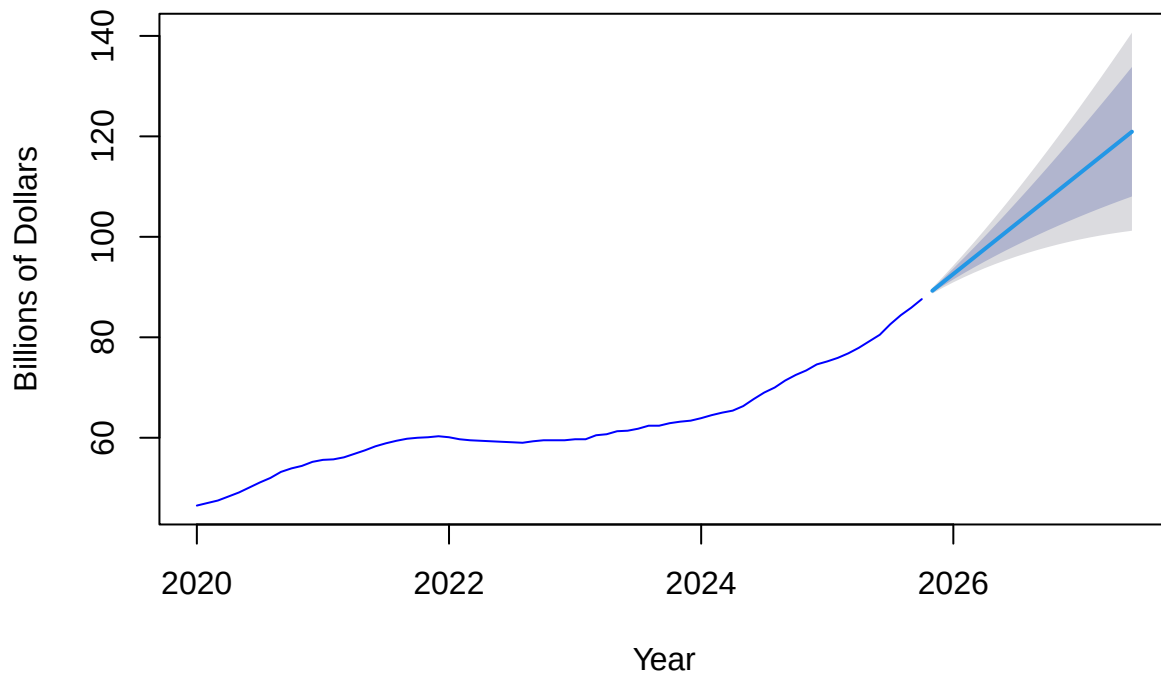
# 5. Forecast the next 20 months
loan_forecast <- forecast(fit, h = 20)

# 6. View and Plot the results
print(loan_forecast)
```

4. Plot

##	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
## Nov 2025	89.26644	88.89909	89.63379	88.70463	89.82825
## Dec 2025	90.93289	90.23112	91.63465	89.85963	92.00614
## Jan 2026	92.59933	91.51354	93.68512	90.93876	94.25990
## Feb 2026	94.26577	92.74897	95.78257	91.94603	96.58551
## Mar 2026	95.93221	93.94123	97.92320	92.88727	98.97716
## Apr 2026	97.59866	95.09360	100.10372	93.76750	101.42981
## May 2026	99.26510	96.20880	102.32139	94.59090	103.93930
## Jun 2026	100.93154	97.28912	104.57396	95.36095	106.50213
## Jul 2026	102.59798	98.33648	106.85949	96.08057	109.11539
## Aug 2026	104.26443	99.35250	109.17635	96.75228	111.77657
## Sep 2026	105.93087	100.33862	111.52312	97.37826	114.48348
## Oct 2026	107.59731	101.29608	113.89854	97.96040	117.23422
## Nov 2026	109.26375	102.22598	116.30152	98.50041	120.02709
## Dec 2026	110.93020	103.12933	118.73106	98.99980	122.86059
## Jan 2027	112.59664	104.00700	121.18628	99.45992	125.73336
## Feb 2027	114.26308	104.85980	123.66636	99.88200	128.64416
## Mar 2027	115.92952	105.68847	126.17058	100.26718	131.59187
## Apr 2027	117.59597	106.49367	128.69826	100.61648	134.57545
## May 2027	119.26241	107.27604	131.24877	100.93085	137.59397
## Jun 2027	120.92885	108.03615	133.82155	101.21117	140.64653

20-Month Forecast: Commercial Bank Real Estate Loans



5. Conclusion

Based on the time-series analysis and the application of the Auto-Regressive Integrated Moving Average (ARIMA) model, the following conclusions can be drawn regarding the commercial bank real estate loan volume:

Forecasted Trend: The model identifies a consistent increasing trend in loan volumes. Over the next 20 months, the total volume is expected to reach approximately \$120.92885 billion by the end of the period.

Statistical Confidence: The forecast provides a high degree of certainty in the short term; however, the widening of the 80% and 95% confidence intervals over the 20-month horizon indicates that external economic volatility remains a significant factor for long-term accuracy.

Market Implication: If the current trajectory continues, the commercial real estate sector will likely see a expansion of credit, suggesting that banking institutions are currently operating under a growth-oriented lending posture relative to historical data.